CS383 · SOFTWARE ENGINEERING

Practice Question Bank Solutions

Practice on the midterm's question types · 8 questions

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01 Solution Q1: ethics

#	Principle	Keyword
1	**CLIENT AND EMPLOYER**	confidential information
2	**PRODUCT**	adequate testing, debugging, and review
3	**JUDGMENT**	deceptive financial practices, bribery
4	**MANAGEMENT**	fair and just remuneration
5	**PROFESSION**	public knowledge of software engineering
6	**COLLEAGUES**	credit the work of others
7	**SELF**	further their knowledge
8	**PUBLIC**	danger to the user, the public, or the environment

Items 4 and 7 are the easiest to get wrong: fair pay is the manager's responsibility (Management), and furthering one's knowledge is the person's own responsibility (Self).

02 Solution Q2: plan-driven or agile

#	Answer	Reason
1	**Plan Driven**	Detailed documentation before coding is a plan-driven trait.
2	**Agile**	The customer is part of the team.
3	**Plan Driven**	The scope and price are fixed, so a full plan is needed from the start.
4	**Agile**	Frequent delivery in short increments.
5	**Agile**	Agile welcomes change.
6	**Plan Driven**	Each phase is finished and approved before the next.

03 Solution Q3: development model

Project	Answer	Justification
1	**Component-Based**	The system is built by assembling ready components: a payment gateway, a package and an SDK. That is the definition of reuse-oriented development.
2	**Plan-Driven**	Safety-critical, requirements fixed by regulation, a large distributed team, documentation and sign-off.
3	**Agile**	Daily feedback, weekly change, 5 co-located developers and a short time to market.
4	**Iterative**	Periodic reviews, a stable core with moderate change, 15 developers, and "regular iterations".

04 Solution Q4: sprint calculations

1) $\text{Sprint Velocity} = \frac{30 + 34 + 28 + 32 + 36}{5} = \frac{160}{5} = 32$

2) $\text{Focus Factor} = \frac{32}{5 \times 8} = \frac{32}{40} = 0.8$

3) Working days = 2 weeks × 5 days = 10 days

   Capacity = 0.8 × (5 × 8 × 10) = 0.8 × 400 = 320 hours

4) 33 ≈ 32 → Feasible

   50 >> 32 → Not feasible

33 is very close to the velocity, so the plan is feasible. 50 is far larger, so the team cannot finish it in one sprint.

05 Solution Q5: User / System and FR / NFR

#	User vs. System	FR vs. NFR	Reason
a	**User**	**NFR**	A general statement, and secure is a quality attribute.
b	**System**	**FR**	Detailed with numbers, and it describes something the system does (lock).
c	**User**	**FR**	A general statement that describes a service (view).
d	**System**	**NFR**	Detailed with numbers, and the subject is speed (performance).

The two questions are independent: User/System is about the level of detail, and FR/NFR is about the content. All four combinations are possible.

06 Solution Q6: Product / Organization / External

#	Answer	Reason
1	**Product**	The availability of the product itself.
2	**Organization**	The company's standard and tools.
3	**External**	A law.
4	**Product**	The product's memory use (efficiency).
5	**Organization**	The development process adopted by the company.
6	**External**	A standard from an outside body.

07 Solution Q7: FR and NFR

Functional
1. Students can search for books by title or author.
2. Students can reserve an available book.
3. Librarians can add new books.
4. Librarians can record returned books.

Non-functional	Kind
1. The system must be available 24 hours a day.	Availability
2. Search results must appear within 3 seconds.	Performance
3. All passwords must be stored encrypted.	Security

08 Solution Q8: short questions

#	Answer
1	Software specification → Software development → Software verification and validation → Software evolution
2	Generic: Microsoft PowerPoint (sold to any customer). Customized: a patient registration system built for one particular hospital.
3	Inflexible to changing requirements · Late discovery of technical problems
4	Process quality · Development technology · People quality · Cost, time and schedule
5	Consistency · Respect · Inclusion · Honesty
6	Inputs: team capacity, product backlog, business conditions, current product, technology. Outputs: sprint goal, sprint backlog.
7	**False**. Maintenance and evolution cost more than development.
8	**False**. The initial estimate is not accurate: the actual value is from `0.25x` to `4x`.
9	Refactoring
10	Benefit: pay as you go, or accessible anywhere. Drawback: lack of control, or security and trust, or internet dependent.